Cransfield Materials Testing Center
Quality Assurance

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## PARTICLE SIZE DISTRIBUTION
## BS 1377 2:1990: 9.2/9.3/9.4 & 9.5

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Client:** | AHP | | | | | **Date submitted:** | 09-Jun-26 |
| **Project/Site Name:** | KIRIAINI AHP | | | | | **Date tested:** | 11-Jun-26 |
| **Sample ID:** | <0.063 | **Sample No.:** | BH01 | **Sample depth (M):** | 0.0 - 25.0 | **Sampled and Submitted by:** | Cransfield |

| **Sample preparation details** | | |
|---|---|---|
| Initial quartered and air dried mass | (gm) | 328 |
| Washed and oven dried mass | (gm) | 176.4 |
| Fine mass | (gm) | 151.6 |
| Fines percent | (%) | 46.2 |

| **Moisture content at sample preparations** | | |
|---|---|---|
| Wet weight of sample | (gm) | 186.5 |
| Dry weight of sample | (gm) | 149.0 |
| Wet weight of water | (gm) | 37.5 |
| Moisture content | (%) | 25.2 |

| **Soil Classification** | | | | | |
|---|---|---|---|---|---|
| **GRAVEL (%):** | 22.8 | **CLAY/SILT (%):** | 71.9 | **USCS** | SILT OF HIGH PLASTICITY(MH) |
| **SAND (%):** | 5.3 | | | **AASHTO** | A-7-5 |

**Wet & Dry Sieve Analysis to BS 1377 2:1990: 9.2/9.3/9.4**

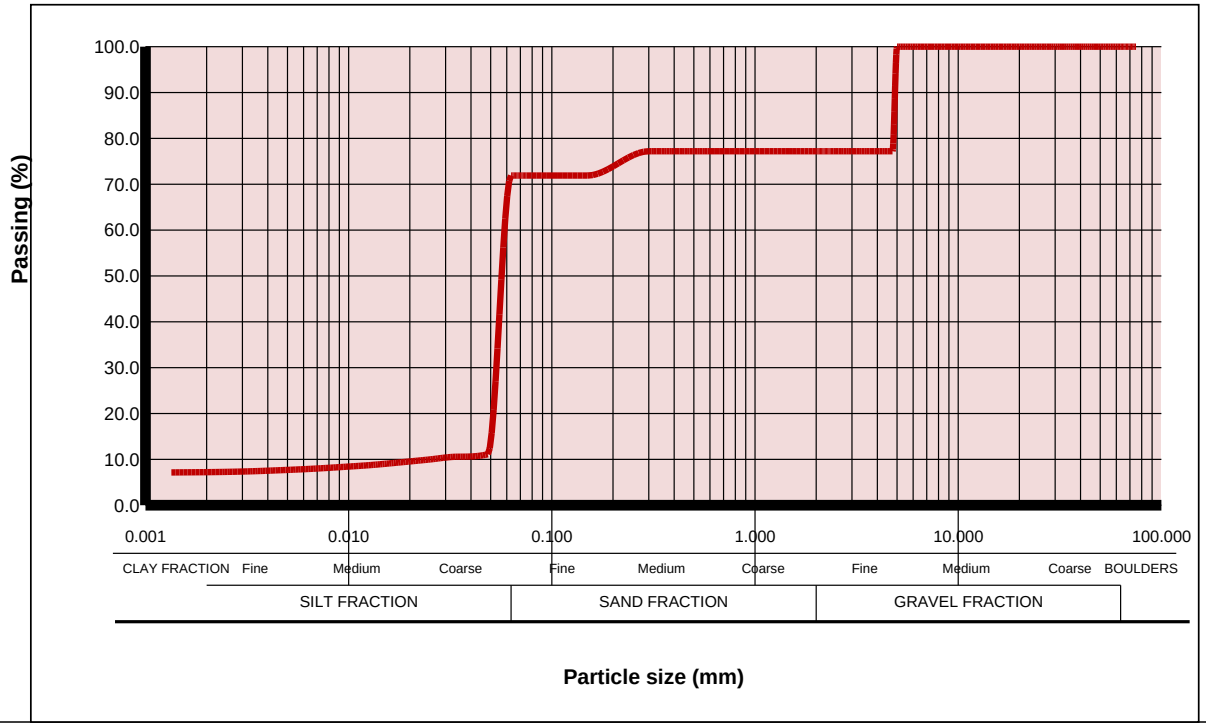
| Sieve size (mm) | Retained mass (gm) | % Retained (%) | Cumulative passing percentage (%) |
|---|---|---|---|
| 75 | 0 | 0.0 | 100.0 |
| 63 | 0 | 0.0 | 100.0 |
| 50 | 0 | 0.0 | 100.0 |
| 37.5 | 0 | 0.0 | 100.0 |
| 28 | 0 | 0.0 | 100.0 |
| 20 | 0 | 0.0 | 100.0 |
| 14 | 0 | 0.0 | 100.0 |
| 10 | 0 | 0.0 | 100.0 |
| 6.3 | 0 | 0.0 | 100.0 |
| 5 | 0 | 0.0 | 100.0 |
| 4.75 (No. 4) | 40.22 | 22.8 | 77.2 |
| 3.35 | 0 | 0.0 | 77.2 |
| 2.36 | 0 | 0.0 | 77.2 |
| 2.00 (No. 10) | 0 | 0.0 | 77.2 |
| 1.18 | 0 | 0.0 | 77.2 |
| 0.6 | 0 | 0.0 | 77.2 |
| 0.425 | 0 | 0.0 | 77.2 |
| 0.3 | 0 | 0.0 | 77.2 |
| 0.15 | 9.35 | 5.3 | 71.9 |
| 0.075 | 0 | 0.0 | 71.9 |
| 0.063 | 0 | 0.0 | 71.9 |
| <0.063 | 126.83 | 71.9 | - |
| TOTAL | 176.4 | - | - |

**Hydrometer Analysis to BS 1377-2:1990:9.5**

| | | | |
|---|---|---|---|
| **Dry Weight** | (g) 176.4 | **Hydrometer Type** | 152H |
| **S.G (Mg/m³)** | 2.65 | **Zero Correction factor** | 0 |
| **Temperature (°C)** | 20 | **Meniscus correction (g/L)** | 0.1 |
| **K Factor** | 1.358 | **Temperature Correction Factor** | |

| Time, min | Actual rh | Adjusted HR(Rh) | Composite correction | Corrected HR | Effective depth (cm) | Diameter; D K√(L/T) | % Fines in Suspension | % Fines based on whole sample |
|---|---|---|---|---|---|---|---|---|
| 0.25 | 24.5 | 24.5 | 0.10 | 24.6 | 12.26 | 0.0956 | 22.4 | 12.0 |
| 0.50 | - | 23.5 | 0.10 | 23.6 | 12.42 | 0.0680 | 21.5 | 11.6 |
| 1 | 22.6 | 22.6 | 0.10 | 22.7 | 12.57 | 0.0484 | 20.7 | 11.1 |
| 2 | 21.5 | 21.5 | 0.10 | 21.6 | 12.75 | 0.0345 | 19.7 | 10.6 |
| 4 | 20.2 | 20.2 | 0.10 | 20.3 | 12.97 | 0.0246 | 18.5 | 9.9 |
| 8 | 18.9 | 18.9 | 0.10 | 19.0 | 13.18 | 0.0175 | 17.3 | 9.3 |
| 15 | 17.8 | 17.8 | 0.10 | 17.9 | 13.36 | 0.0129 | 16.3 | 8.8 |
| 30 | 16.9 | 16.9 | 0.10 | 17.0 | 13.51 | 0.0092 | 15.5 | 8.3 |
| 60 | 16.1 | 16.1 | 0.10 | 16.2 | 13.64 | 0.0065 | 14.7 | 7.9 |
| 120 | 15.5 | 15.5 | 0.10 | 15.6 | 13.74 | 0.0046 | 14.2 | 7.6 |
| 240 | 15.0 | 15.0 | 0.10 | 15.1 | 13.82 | 0.0033 | 13.7 | 7.4 |
| 480 | 14.7 | 14.7 | 0.10 | 14.8 | 13.87 | 0.0023 | 13.5 | 7.2 |
| 1440 | 14.5 | 14.5 | 0.10 | 14.6 | 13.90 | 0.0013 | 13.3 | 7.1 |

### PARTICLE SIZE DISTRIBUTION GRAPH




| **Tested by:** | JILLO | **Date reported:** | _____ | Checked by: _____ |
|---|---|---|---|---|